Task-1 on AWS_EC2

1. **Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.**

Steps:

1.**Open the AWS EC2 Console** and click **Launch Instance**

2.**Name and tags:** Given your server name as "**first-server**"

3.Application and OS Images (AMI) Click on the **Amazon Linux**

4.**Instance Type:** Select `t2.micro` or `t3.micro` to keep it free-tier

5.**Key Pair:** Selected my existing key pair "**linux-testing**"

6.Network Settings : Make sure **Allow HTTP traffic from the internet (Port 80)**

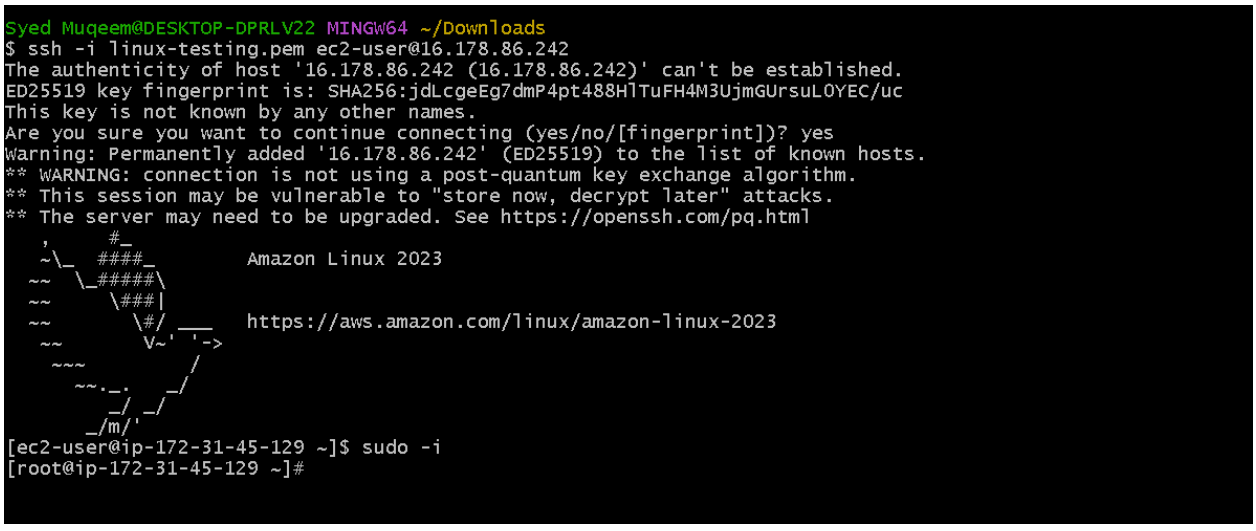
7. **To write the Script**:

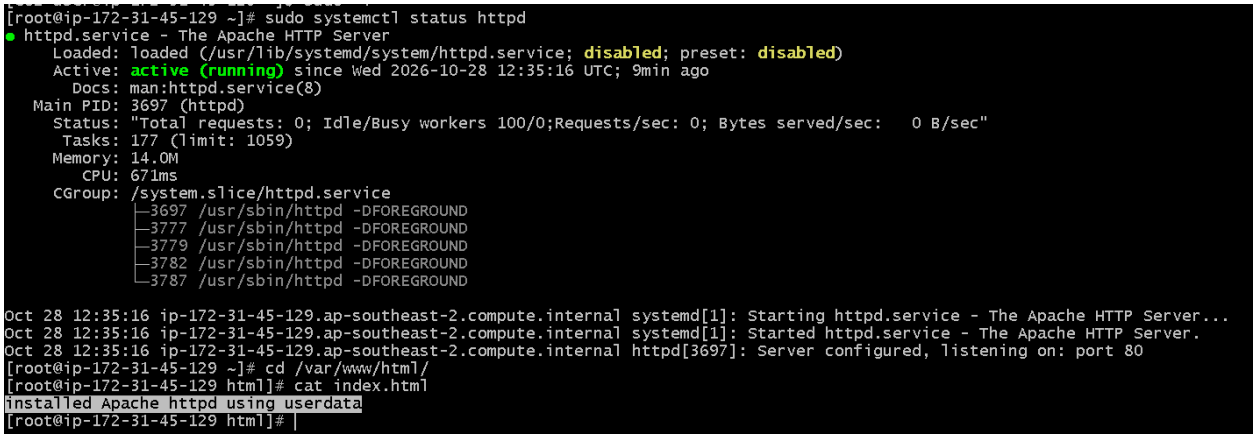
Scroll to the very bottom and expand the **Advanced Details** section

Scroll down to the final multi-line field labeled **User data**

Write the script you want: ex: Install httpd

8.Click **Launch Instance**





2. **Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.**

Steps:

1.Open the AWS EC2 Console and click **Launch Instance**

2.Name and tags**:** Given your server name as “**2nd-server**”

3.Application and OS Images (AMI) Click on the **Ubuntu**

4.**Instance Type:** Select `t2.micro` or `t3.micro` to keep it free-tier

5.**Key Pair:** Selected my existing key pair “**linux-testing**”

6.Network Settings : I have used existing group

7. **To write the Script**:

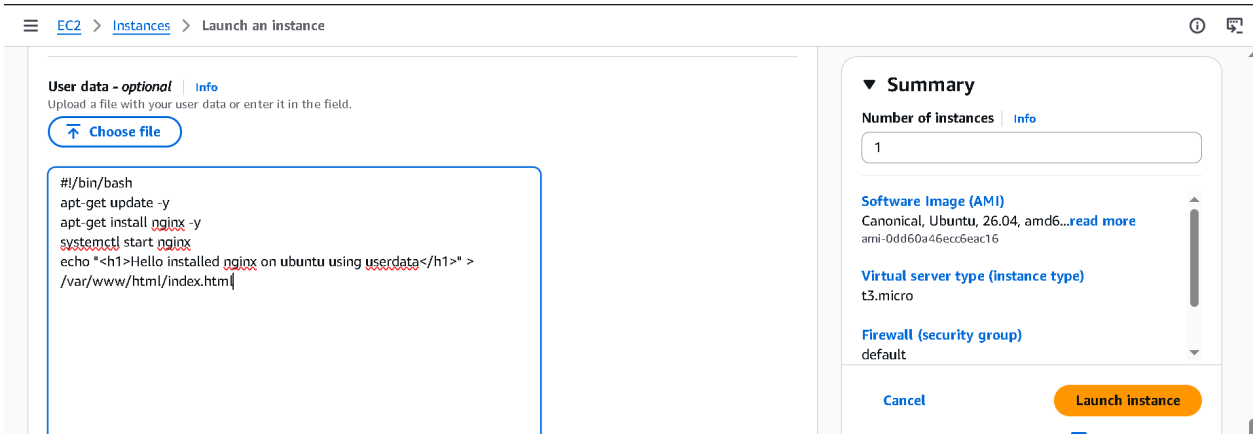
Scroll to the very bottom and expand the **Advanced Details** section

Scroll down to the final multi-line field labeled **User data**

Write the script you want: ex: to install the nginx on ubuntu

8.Click **Launch Instance**

9.connect to ec2-instance using ssh



ubuntu@ip-172-31-33-183: ~

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/Downloads
$ ssh -i "linux-testing.pem" ubuntu@ec2-54-66-142-0.ap-southeast-2.compute.amazonaws.com
The authenticity of host 'ec2-54-66-142-0.ap-southeast-2.compute.amazonaws.com (54.66.142.0)' can't be established.
ED25519 key fingerprint is: SHA256:byoKssTvkLJSc08SO3iOhxjXS0BeeGnJkOi1T14d9T8
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-66-142-0.ap-southeast-2.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Oct 28 13:19:47 UTC 2026

  System load:  0.19              Temperature:           -273.1 C
  Usage of /:   33.9% of 6.61GB   Processes:             124
  Memory usage: 27%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.33.183

Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



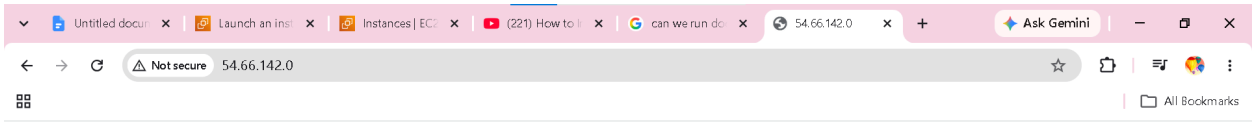
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-33-183:~$
```

```
ubuntu@ip-172-31-33-183:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-28 13:18:13 UTC; 5min ago
 Invocation: c1ce8b338ff14b7383ed75882f0bb7b8
       Docs: man:nginx(8)
    Process: 1424 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1426 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1452 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3.1M (peak: 7M)
        CPU: 42ms
     CGroup: /system.slice/nginx.service
             ├─1452 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─1454 "nginx: worker process"
             └─1455 "nginx: worker process"

Oct 28 13:18:12 ip-172-31-33-183 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 28 13:18:13 ip-172-31-33-183 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-33-183:~$
```



Hello installed nginx on ubuntu using userdata

3. **Launch one EC2 using Windows and install Tomcat on Windows.**

Steps:

1.Open the AWS EC2 Console and click **Launch Instance**

2.Name and tags**:** Given your server name as "**windows-server**"

3.Application and OS Images (AMI) Click on the **Windows**

4.**Instance Type:** Select `t3.micro` to keep it free-tier

5.**Key Pair:** Selected my existing key pair "**linux-testing**"

6.Network Settings : I have used existing group

7. **write the Script**:

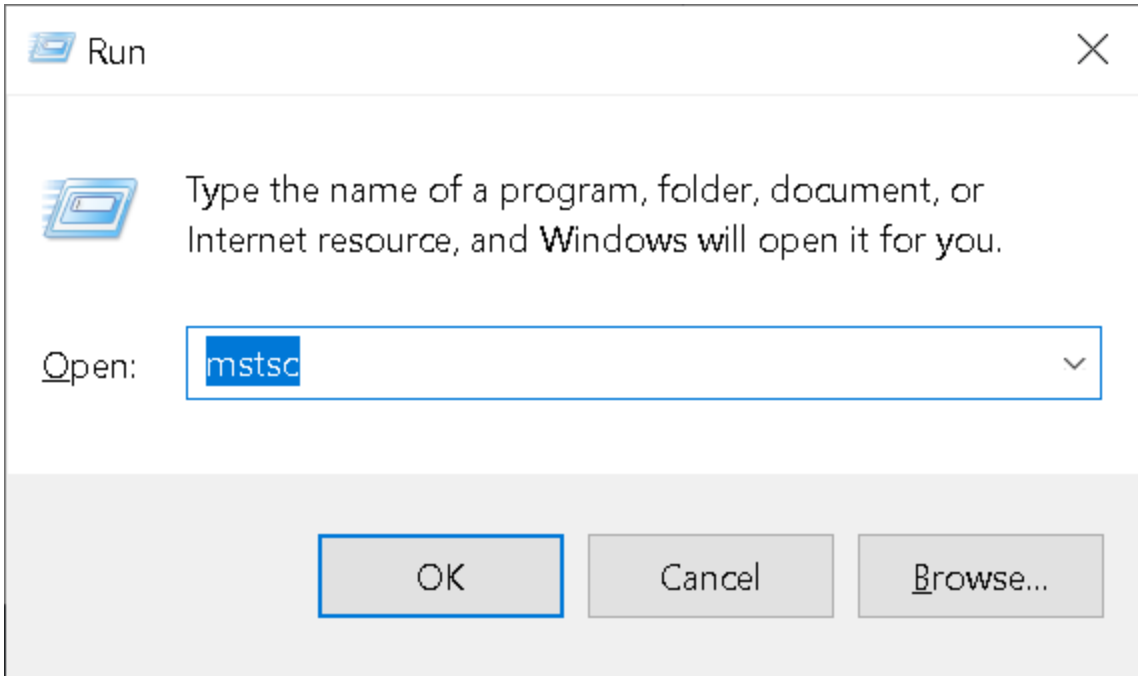
Scroll to the very bottom and expand the **Advanced Details** section

Scroll down to the final multi-line field labeled **User data**

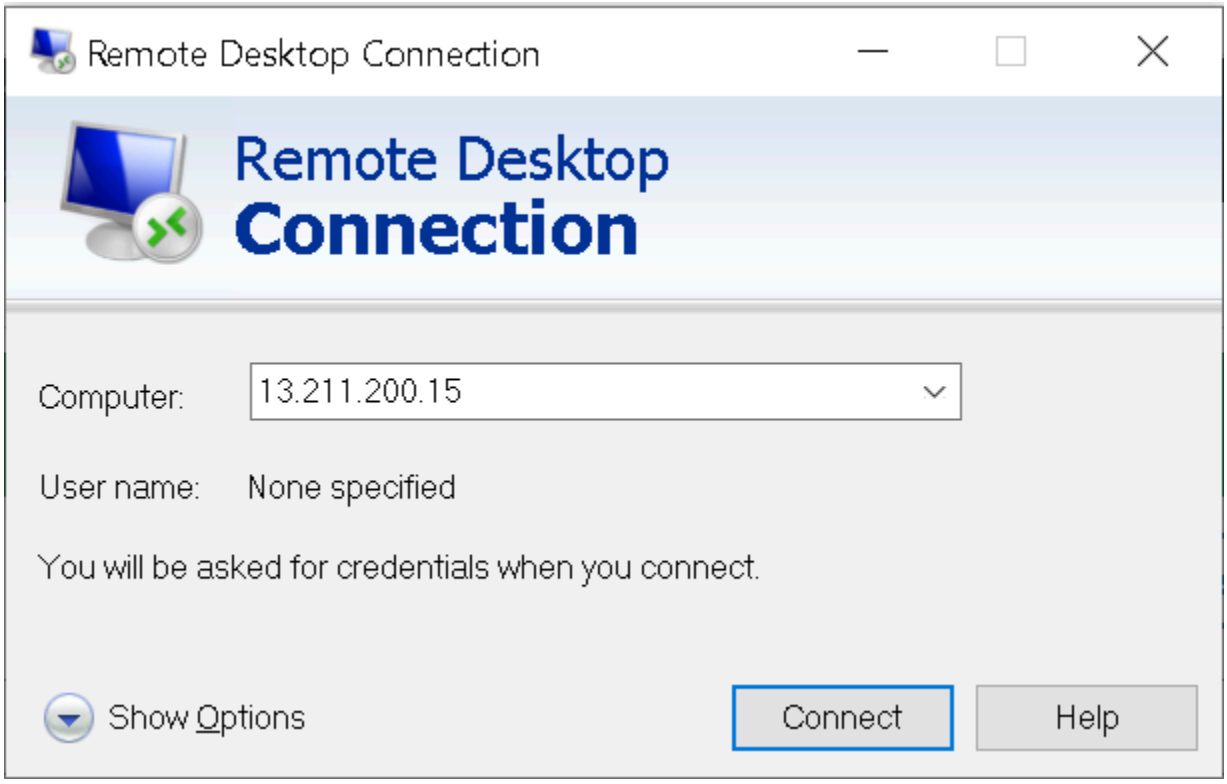
Write the script : to **install the Tomcat on windows**

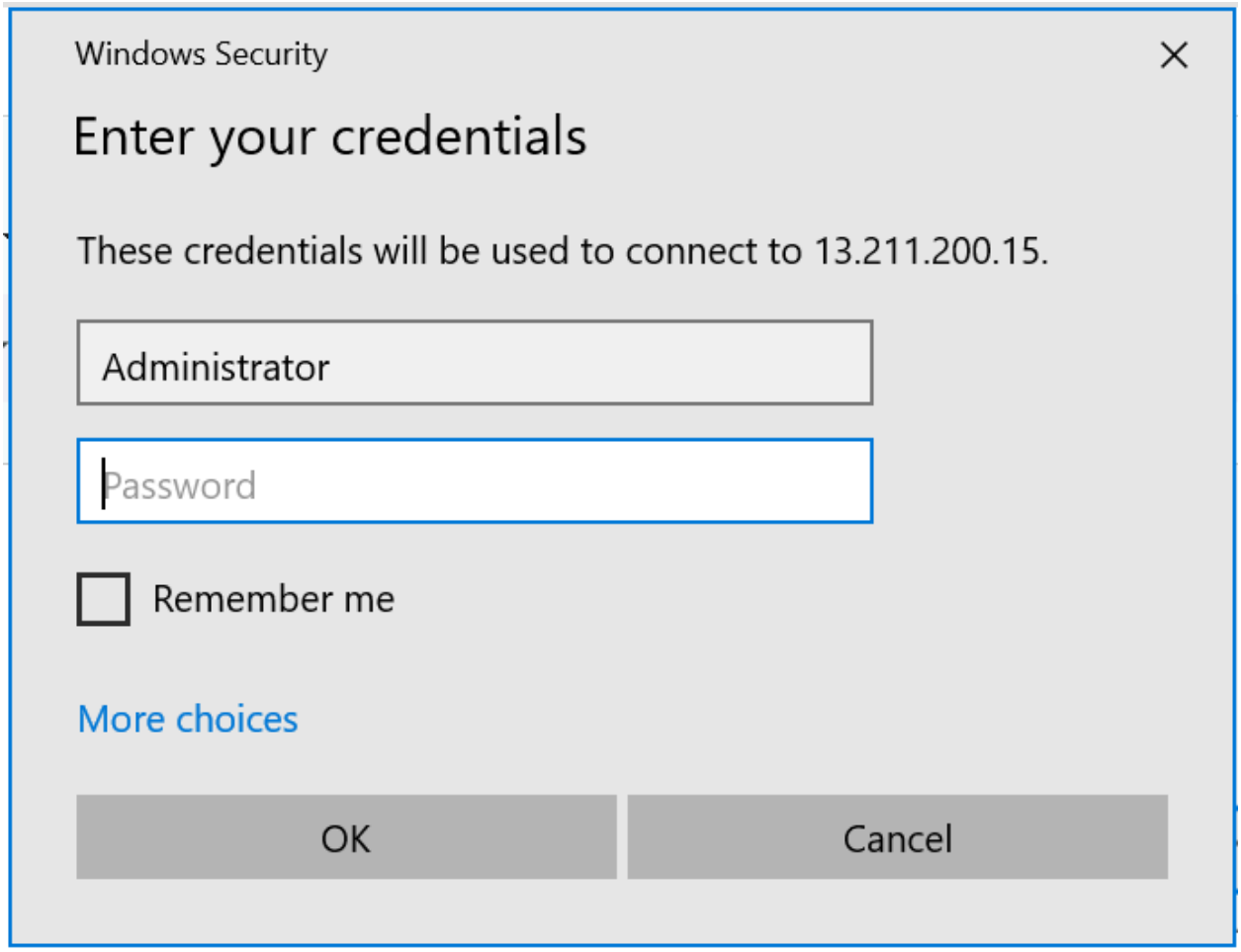
8.Click **Launch Instance**

9. After the lunch of instance copy the public ip and in "widows run' type mstsc and enter



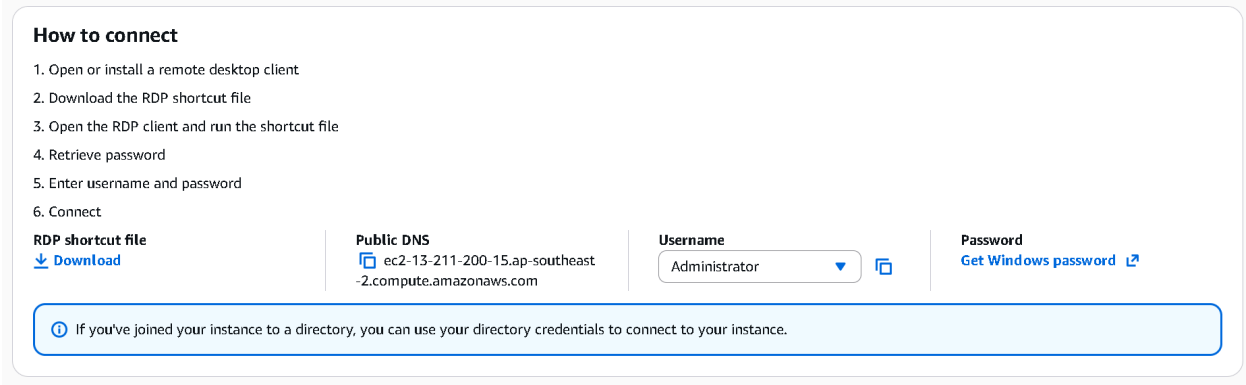
10. Paste the public ip copied from the instance and click on connect.



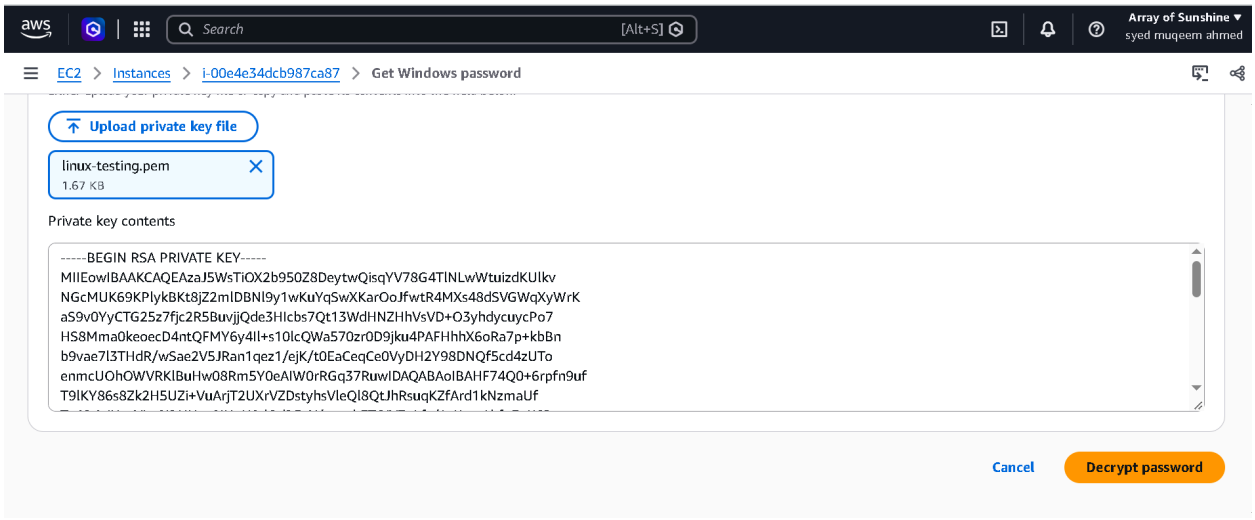


11. For username and password got to ec2 instance click on connect option

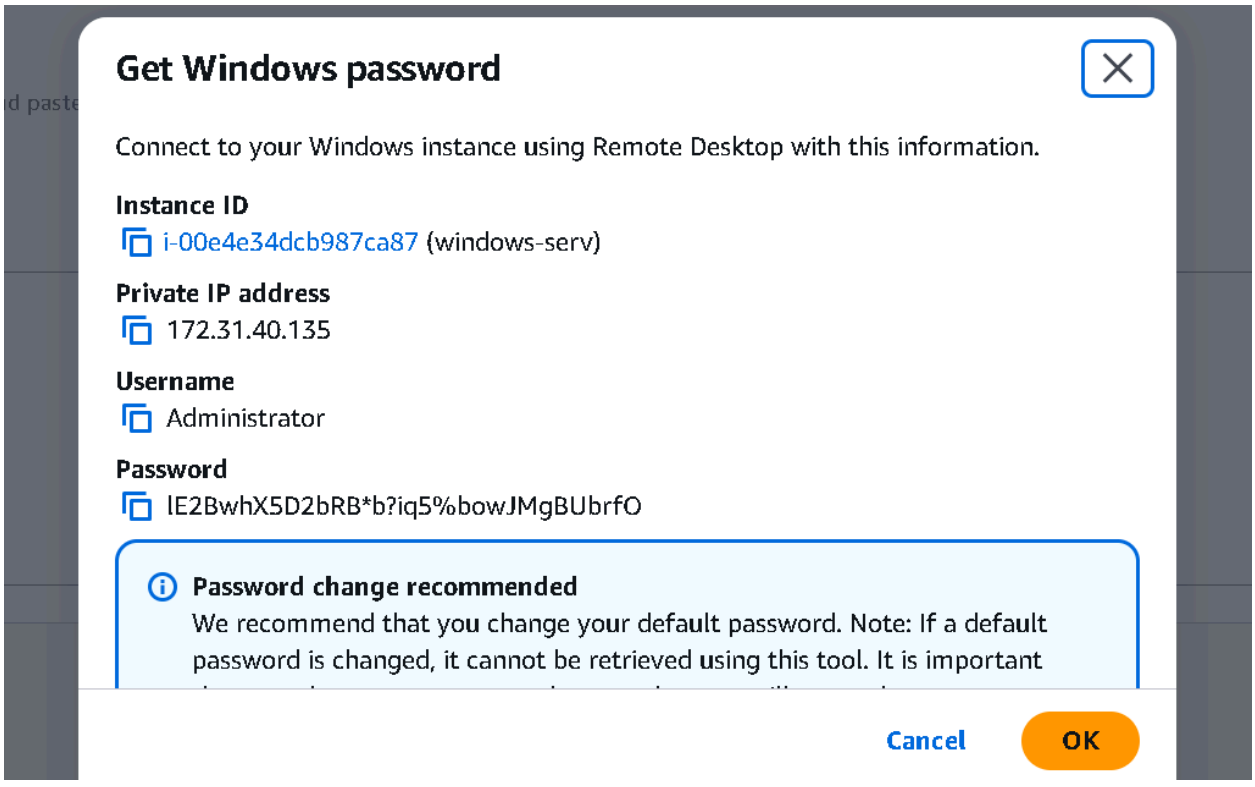
And click on RDP (remote desktop protocol) at the bottom you can see



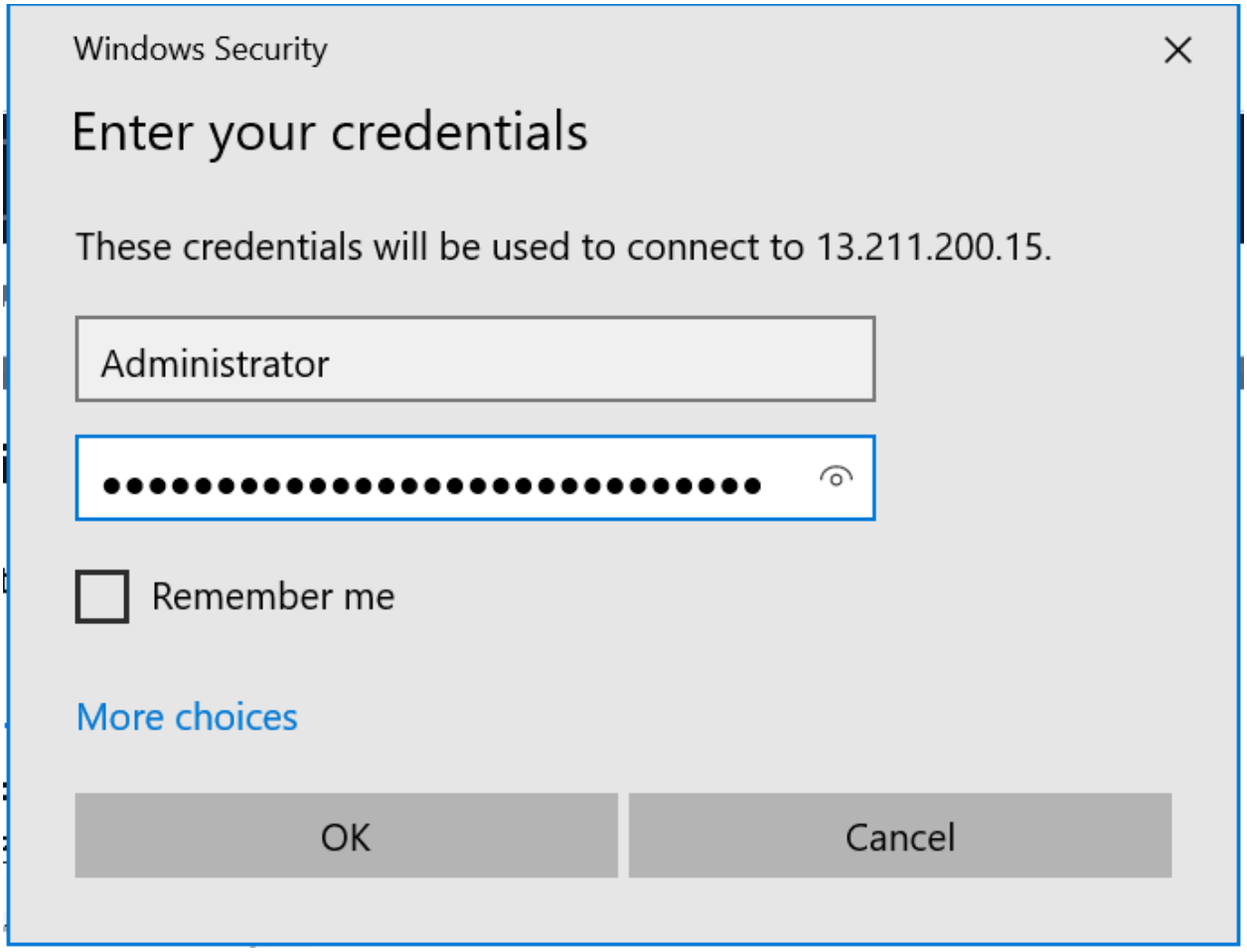
12. Download RDP shortcut file, copy username provided as administrator and click on Get windows password.



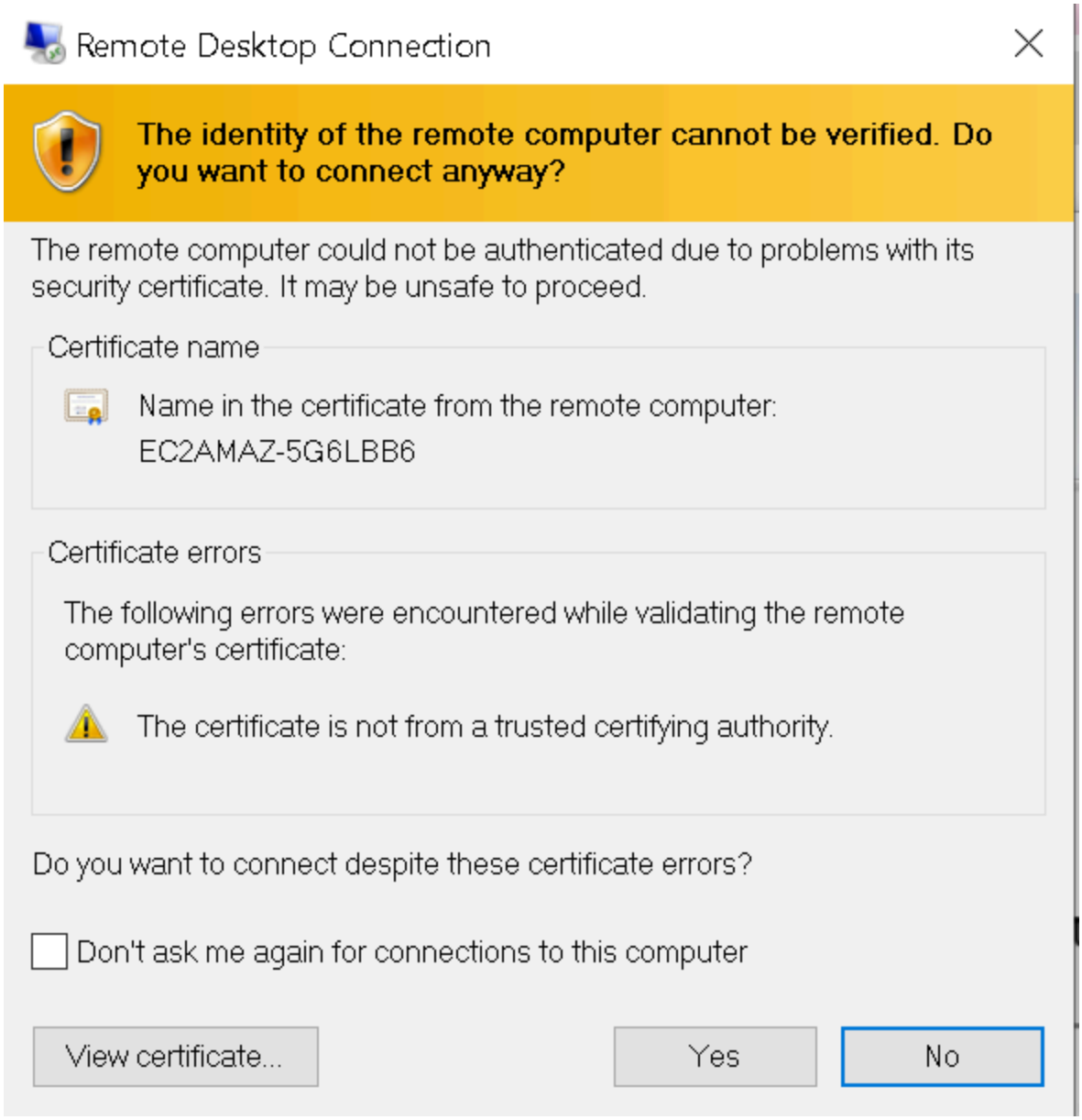
13. Upload your .pem key and click on **Decrypt password**

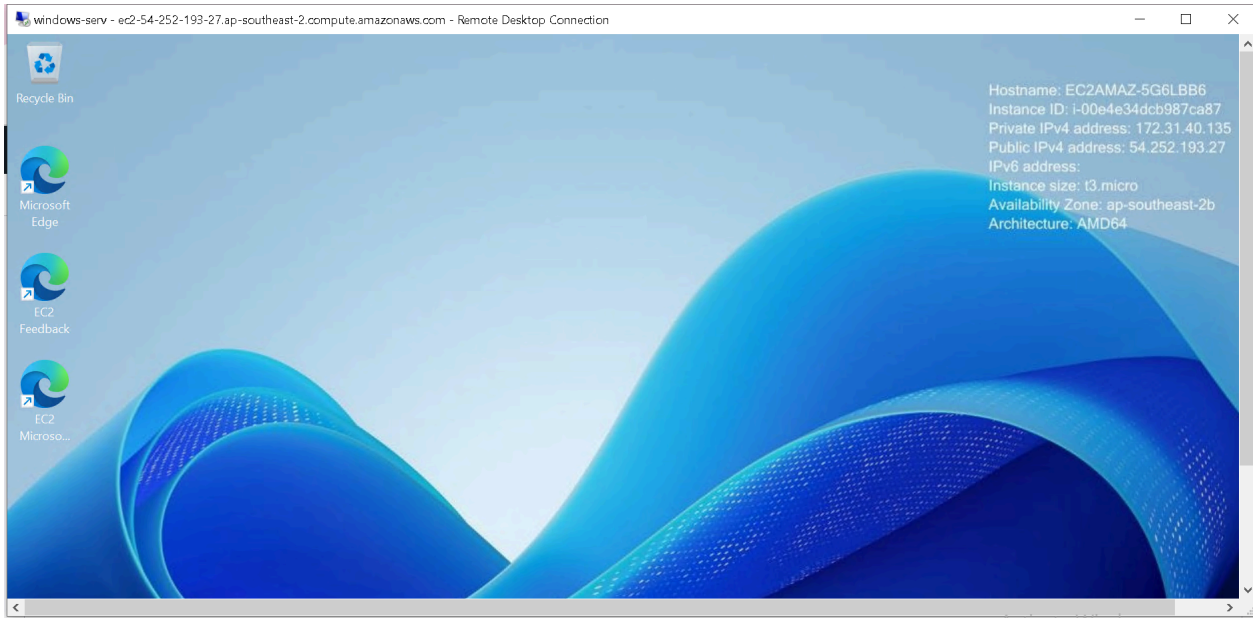


14. Copy the password and paste it in the password field and click ok.


Windows Security
Enter your credentials
These credentials will be used to connect to 13.211.200.15.
Administrator
Remember me
More choices
OK
Cancel

Click yes

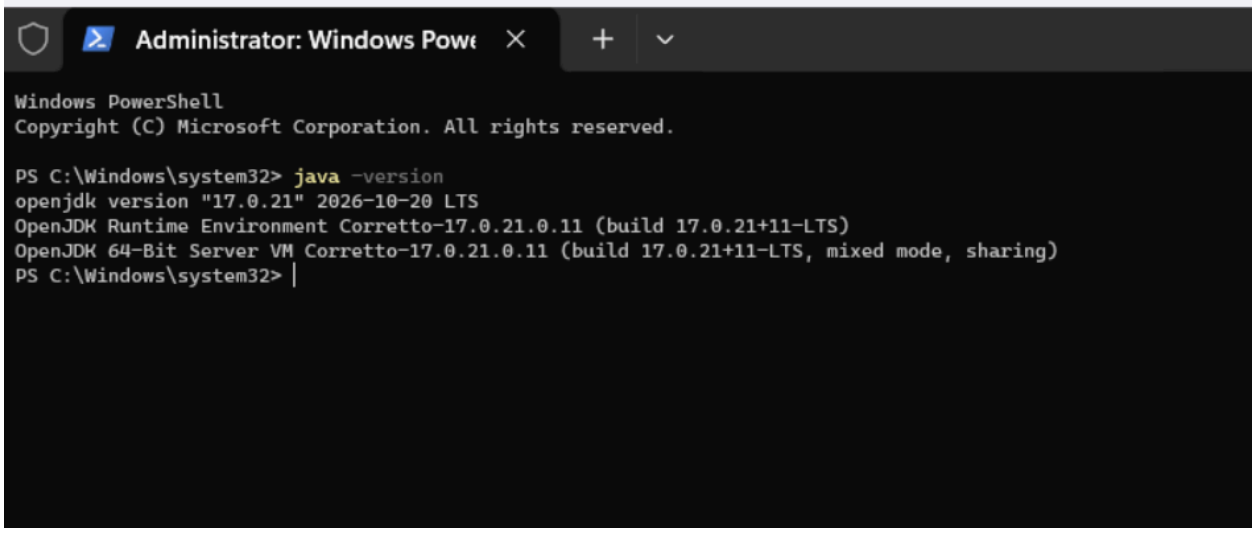

Remote Desktop Connection
The identity of the remote computer cannot be verified. Do you want to connect anyway?
The remote computer could not be authenticated due to problems with its security certificate. It may be unsafe to proceed.
Certificate name
Name in the certificate from the remote computer:
EC2AMAZ-5G6LBB6
Certificate errors
The following errors were encountered while validating the remote computer's certificate:
The certificate is not from a trusted certifying authority.
Do you want to connect despite these certificate errors?
Don't ask me again for connections to this computer
View certificate...
Yes
No

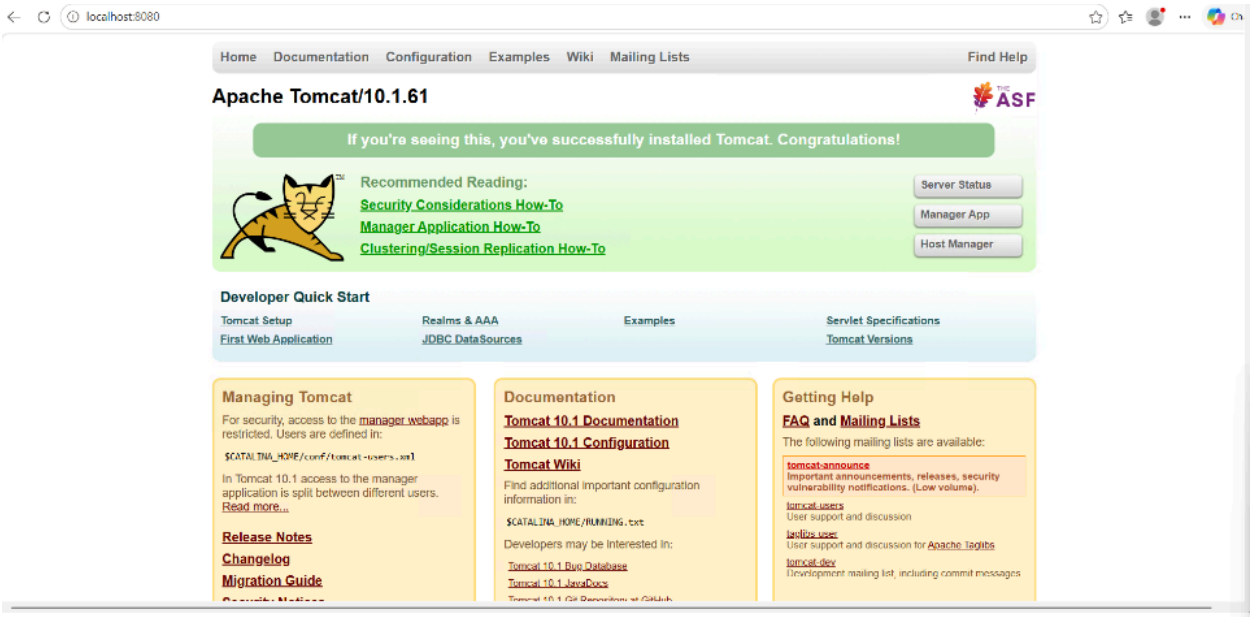


To install tomcat in windows follow the belo steps.

Steps:

1. Open windows powershell
2. Java -version





4. Take a snapshot of the instance created in Task 1.

Steps:

1. Open AWS Console → EC2 → Instances.

2. Select the Amazon Linux 2 EC2 instance that you created with the Apache User Data script.

3. Make sure the instance is Running and the status checks are complete.

4. Click Actions → Image and templates → Create image.

5. Enter an image name, for example: windows-Server-Backup

6. Add a description if required.

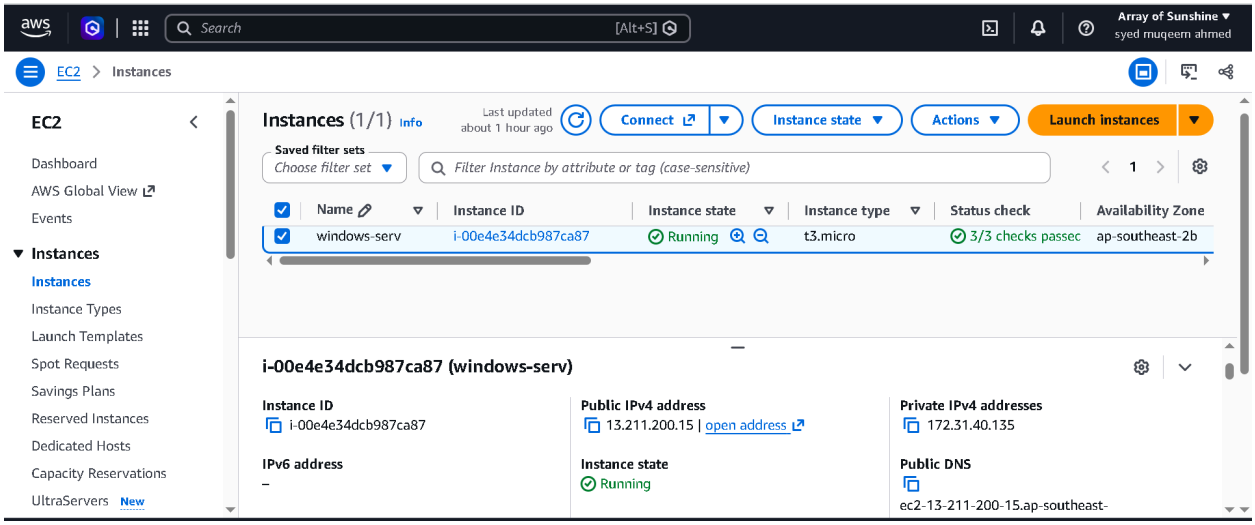
7. Keep the default volume settings.

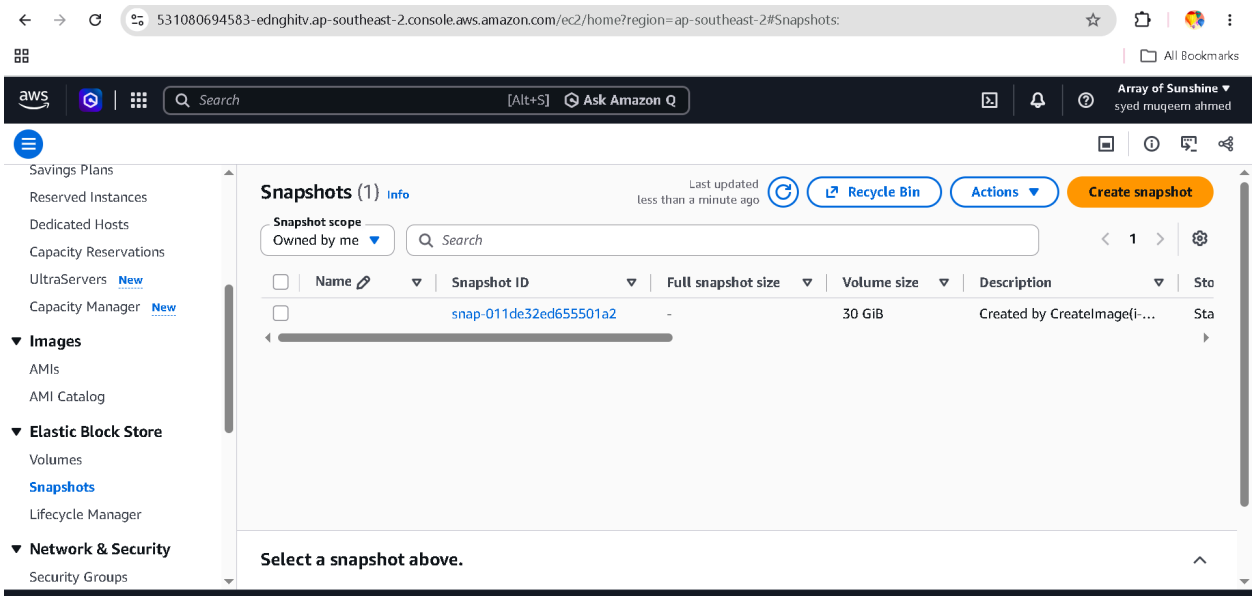
8. Click Create image.

9. Go to EC2 → AMIs and verify that the AMI is being created.

10. Go to EC2 → Snapshots and verify the snapshot created from the instance's EBS volume.

11. Wait until the snapshot status becomes Completed.





5. Assign passwordless authentication for the EC2 created in Task 2.

Step 1: Generate SSH Key Pair in Git Bash

Step 2: Display the Public Key

Step 3: Connect to EC2 Using the Existing PEM Key

Step 4: Create the SSH Directory

Step 5: Add the Public Key to EC2

Step 6: Set File Permissions

Step 7: Exit the EC2 Instance

Step 8: Test Passwordless Authentication

ubuntu@ip-172-31-38-74: ~

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/Downloads
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOp+ftr36lvwPsLmNAxDfMTO5nLHvdSx7GJSWx1LZqqG Syed Muqeem@DESKTOP-DPRLV22

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/Downloads
$ ssh -i "linux-testing.pem" ubuntu@ec2-3-25-212-196.ap-southeast-2.compute.amazonaws.com
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Oct 29 11:40:09 UTC 2026

  System load:  0.08              Temperature:           -273.1 C
  Usage of /:   33.9% of 6.61GB   Processes:             119
  Memory usage: 27%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.38.74


Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu Oct 29 11:06:07 2026 from 103.203.172.230
ubuntu@ip-172-31-38-74:~$ chmod 400 "/c/Users/YourName/Downloads/your-key.pem"
chmod: cannot access '/c/Users/YourName/Downloads/your-key.pem': No such file or directory
ubuntu@ip-172-31-38-74:~$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
nano ~/.ssh/authorized_keys
ubuntu@ip-172-31-38-74:~$ nano ~/.ssh/authorized_keys
ubuntu@ip-172-31-38-74:~$ chmod 600 ~/.ssh/authorized_keys
ubuntu@ip-172-31-38-74:~$ exit
logout
Connection to ec2-3-25-212-196.ap-southeast-2.compute.amazonaws.com closed.
```

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/Downloads
$ ssh -i ~/.ssh/id_ed25519 ubuntu@3.25.212.196
The authenticity of host '3.25.212.196 (3.25.212.196)' can't be established.
ED25519 key fingerprint is: SHA256:+vs3dpEiPy5m/6aKRUfa/le3XpUzu2/DOvbQ1PHYXtk
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:44: ec2-3-25-212-196.ap-southeast-2.compute.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.25.212.196' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Oct 29 11:45:45 UTC 2026

  System load:  0.02              Temperature:           -273.1 C
  Usage of /:   33.9% of 6.61GB   Processes:             118
  Memory usage: 27%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.38.74


Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu Oct 29 11:40:10 2026 from 103.203.172.230
ubuntu@ip-172-31-38-74:~$
```

6. Launch any EC2 using the spot purchasing option.

Steps:

1. Open AWS Console: Go to the EC2 Dashboard.

2. Click Launch instance: Select Launch instance.

3. Enter instance details:

Ex - Name: Spot-Instance

Ex - AMI: Ubuntu Server

- Instance type: t3.micro (if available in your region)

- Key pair: Select an existing key pair or create a new one.

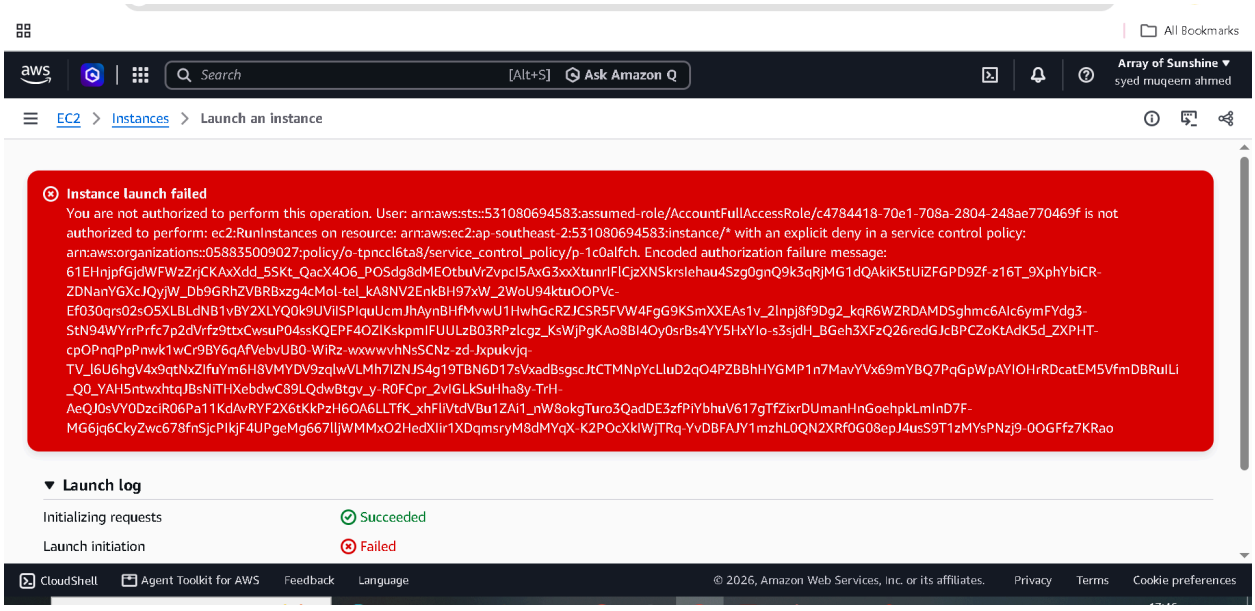
4. Configure network settings: Select a VPC and subnet. Allow SSH (port 22) from your IP if you need SSH access.

5. Choose Spot purchasing option: Expand Advanced details, find Purchasing option, and select Spot.

6. Review settings: Check the instance configuration and estimated pricing.

7. Launch the instance: Click Launch instance and wait for AWS to fulfill the Spot request.

8. Verify: Open EC2 → Instances, select your instance, and check that its purchasing option or lifecycle indicates Spot.

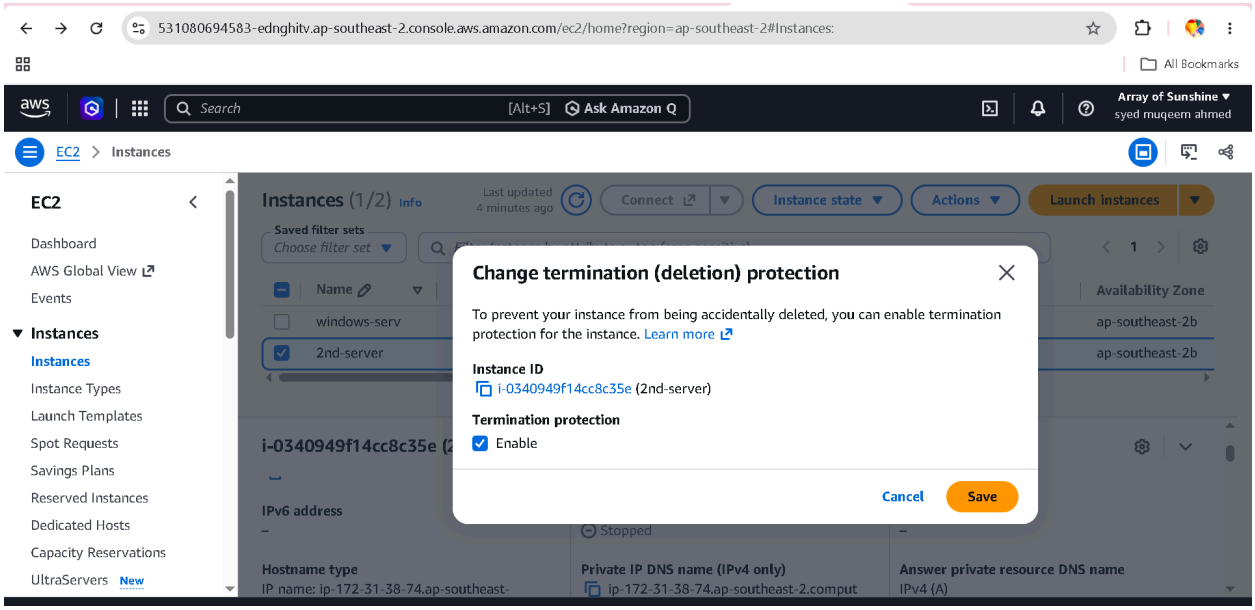


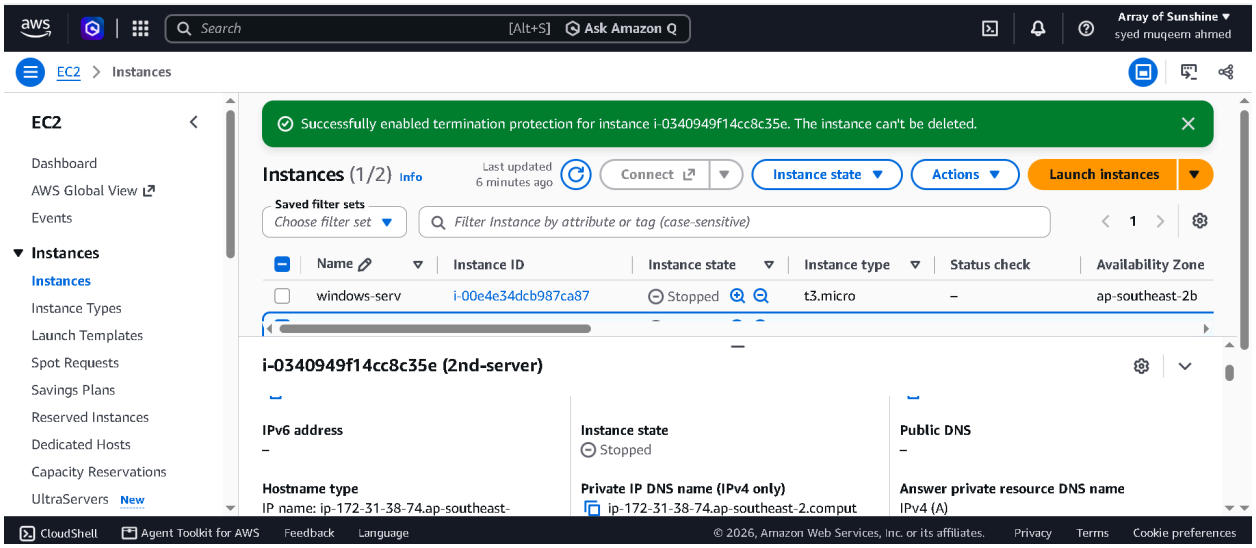
7. Enable termination policy on the EC2 created in Task 2.

Steps:

1. Open the AWS EC2 Console.
2. In the left menu, click Instances.
3. Select your EC2 instance.
4. Click Actions → Instance settings → Change termination protection.
5. Select Enable.
6. Click Save or Apply to confirm.

Important: AWS does not allow termination protection on Spot Instances. If your instance uses Spot purchasing, this option will not be available. Use a regular On-Demand EC2 instance instead.





8. Launch one EC2 using AWS CLI.

Task: Launch an EC2 Instance Using AWS CLI

1. Open Git Bash on Windows.

2. Check whether AWS CLI is installed.

3. Configure AWS CLI using your AWS Learner Lab credentials.

4. Set the default AWS region to Asia Pacific (Sydney), `ap-southeast-2`.

5. Verify AWS credentials and account identity.

6. Check the available EC2 key pairs and select `linux-testing`.

7. Search for the latest available Ubuntu 24.04 AMI from Canonical.

8. Copy the AMI ID returned by AWS.

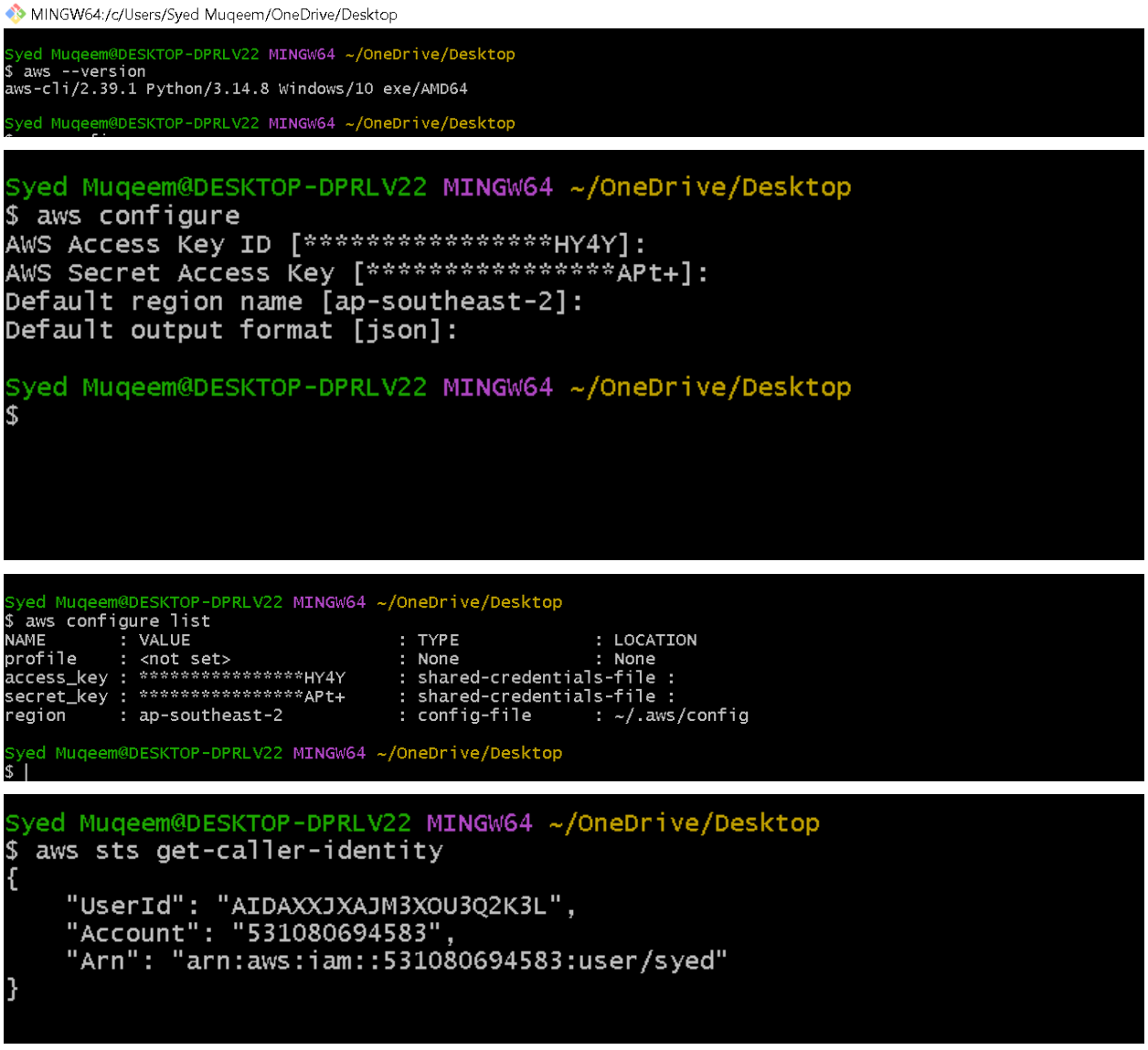
9. Launch one EC2 instance using the Ubuntu AMI, `t3.micro` instance type, and `linux-testing` key pair.

10. Assign the instance name `CLI-Ubuntu`.

11. Verify the instance ID, instance type, private IP address, and current state.

12. Confirm that the instance state changes from `pending` to `running`.

13. Open the AWS EC2 Console and verify that the `CLI-Ubuntu` instance appears in the instance list.



```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-key-pairs --query "KeyPairs[].KeyName" --output table
-------------------
|DescribeKeyPairs |
+-----------------+
|  linux-testing  |
+-----------------+
```

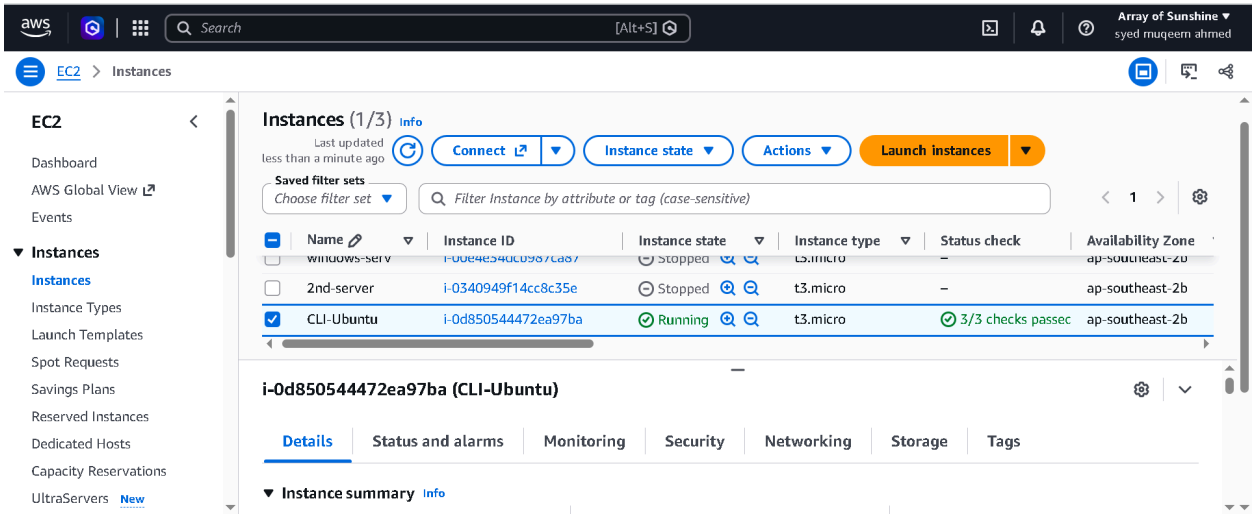
```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-images \
  --owners 099720109477 \
  --filters "Name=name,Values=ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-*" \
            "Name=state,Values=available" \
  --query "Images | sort_by(@, &CreationDate)[-1].[ImageId,Name]" \
  --output table
-------------------------------------------------------------------------
|                          DescribeImages                               |
+-----------------------------------------------------------------------+
|  ami-01e91e94accc42eec                                                |
|  ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20261023   |
+-----------------------------------------------------------------------+


Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop
$ aws ec2 run-instances \
  --image-id ami-01e91e94accc42eec \
  --instance-type t3.micro \
  --key-name linux-testing \
  --count 1 \
  --tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=CLI-Ubuntu}]'
{
    "ReservationId": "r-0c3bc1d8d1e0425ee",
    "OwnerId": "531080694583",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "10c81d22-d058-4d6d-846c-b16488808e83",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2026-10-29T13:38:55+00:00",
                        "AttachmentId": "eni-attach-0ccb9b8f8f25fd77b",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
                        "Status": "attaching",
                        "NetworkCardIndex": 0
                    },
                    "Description": "",
                    "Groups": [
                        {
```

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-instances \
  --filters "Name=tag:Name,Values=CLI-Ubuntu" \
  --query "Reservations[].Instances[].[InstanceId,State.Name,InstanceType,PrivateIpAddress]" \
  --output table
-----------------------------------------------------------------
|                       DescribeInstances                       |
+----------------------+----------+-----------+-----------------+
|  i-0d850544472ea97ba |  running |  t3.micro |  172.31.45.205  |
+----------------------+----------+-----------+-----------------+


Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop
$ aws ec2 describe-instances \
  --filters "Name=tag:Name,Values=CLI-Ubuntu" \
  --query "Reservations[].Instances[].State.Name" \
  --output text
running


Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop
$
```



Result: Successfully launched an Ubuntu EC2 instance using AWS CLI from Git Bash.